

Workforce Ecosystem 360: Decoding Sourcing Channel Efficiency, Operational Productivity, Absenteeism Patterns, and Predictive Attrition

1. Project Overview

This project involves cleaning, transforming, and analyzing raw Human Resources (HR) workforce datasets using Python followed by constructing an interactive multi-page Power BI dashboard to derive actionable workforce insights.

Objective

The main objective is to evaluate talent acquisition channel efficiency, monitor operational productivity across work modes and departments, detect absenteeism/tardiness anomaly patterns, and predict flight risk to minimize employee turnover.

2. Data Sources (Sample)

- **Source Description and Timeline:** Open Source and 2018 to 2026.
 - **Domain:** Human Resources & People Analytics.
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3. Problem Statement (Sample)

- **Sourcing Optimization:** To evaluate hiring channel performance (Naukri, LinkedIn, Referrals, etc.) in terms of cost per hire, volume, and post-hire attrition.
 - **Productivity & Training Yield:** To assess employee productivity and KPI achievement rates across different work environments (WFH, Onsite, Hybrid) and measure training effectiveness.
 - **Flight Risk & Absenteeism:** To analyze early disengagement indicators (late arrivals, sick/casual leaves, unplanned absences) and build a predictive retention strategy strategies.
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4. Attribute (Column /Features) Details:

Dataset Column/features Description:

- **Employee_ID:** Unique identification code assigned to each employee
- **Employee_Name:** Full name of the employee
- **Manager_Employee_ID:** Employee ID of the reporting manager to define organizational hierarchy
- **Gender:** Demographic gender identification of the employee
- **Date_Of_Birth:** Employee's birth date used to calculate age groups
- **Mail_ID:** Official company email address of the employee

- Department: Functional business unit (e.g., HR, Technology, Sales, Operations, Finance)
 - Designation: Employee job tier and role level within the company hierarchy
 - Region_Location: Geographical office location or work branch
 - Source_Of_Hiring: Recruitment channel used to acquire the candidate (e.g., Naukri, LinkedIn, Referral)
 - Cost_Per_Hire: Financial expenditure incurred to acquire the employee
 - Experience_Type: Candidate entry level (e.g., Fresher, Experienced, Internal Mobility)
 - Date_Of_Joining: Official employment start date
 - Last_Working_Date: Exit date recorded for resigned or terminated employees
 - Total_Working_Days: Total standard working days in the evaluation period
 - Days_Present: Total days the employee was physically or remotely present for work
 - Sick_Leaves_Taken: Number of authorized sick leave days utilized
 - Casual_Leaves_Taken: Number of authorized casual leave days utilized
 - Unplanned_Absences: Number of emergency or unexcused absences taken without prior notice
 - Late_Arrival_Count: Frequency of logging in after standard shift start time
 - Reason_For_Leaving: Primary cause stated for employee departure
 - Employment_Status: Current work status of the individual (e.g., Active, Resigned, Terminated)
 - Work_Mode: Primary work arrangement (e.g., Onsite, Hybrid, Remote)
 - Transportation_Type: Primary mode of commute used by the employee
 - Annual_CTC: Total annual compensation package (Cost to Company)
 - Training_Hours: Total hours spent in professional development and skill training programs
 - Overtime_Hours: Total extra working hours logged beyond standard shift hours
 - Productivity_Score: Measured score representing employee work output efficiency
 - KPI_Achievement_Rate: Percentage of Key Performance Indicator targets successfully met
 - Engagement_Score: Overall internal survey score measuring workplace engagement
 - Satisfaction_Score: Job satisfaction score rated on a standard scale
 - Flight_Risk_Indicator: Predicted likelihood rating of an employee leaving the company
 - Performance_Rating: Overall score received during the annual performance appraisal
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5. Tools & Technologies

- **Python & Google Colab:** Initial data cleaning, missing value handling, type casting, and exploratory analytics..
 - **Power BI:** Data modelling, DAX calculations, visualization, and interactive dashboard creation.
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6. Data Pre-Processing (Python)

Tasks Performed:

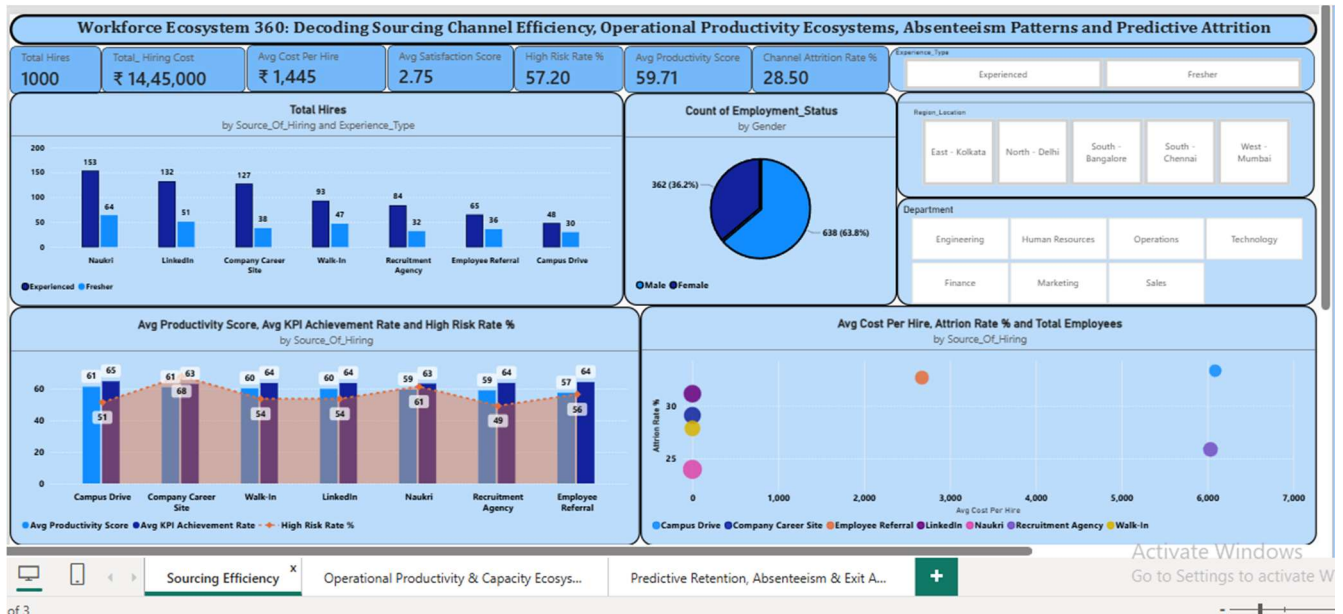
- **Data Cleaning & Transformation:** Handled missing values in leave records, standardized categorical text columns, converted data types (dates, numerical metrics, currency formats), and calculated cost-per-hire fields
 - **Feature Engineering:** Created derived fields such as Flight_Risk_Indicator based on attendance anomalies and calculated Cost per Hire.
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7. DAX (Power BI)

- **Calculated Columns & DAX Measures:** Implemented DAX formulas for key metrics, such as total sales, growth percentages, and cumulative totals. **(List Out)**

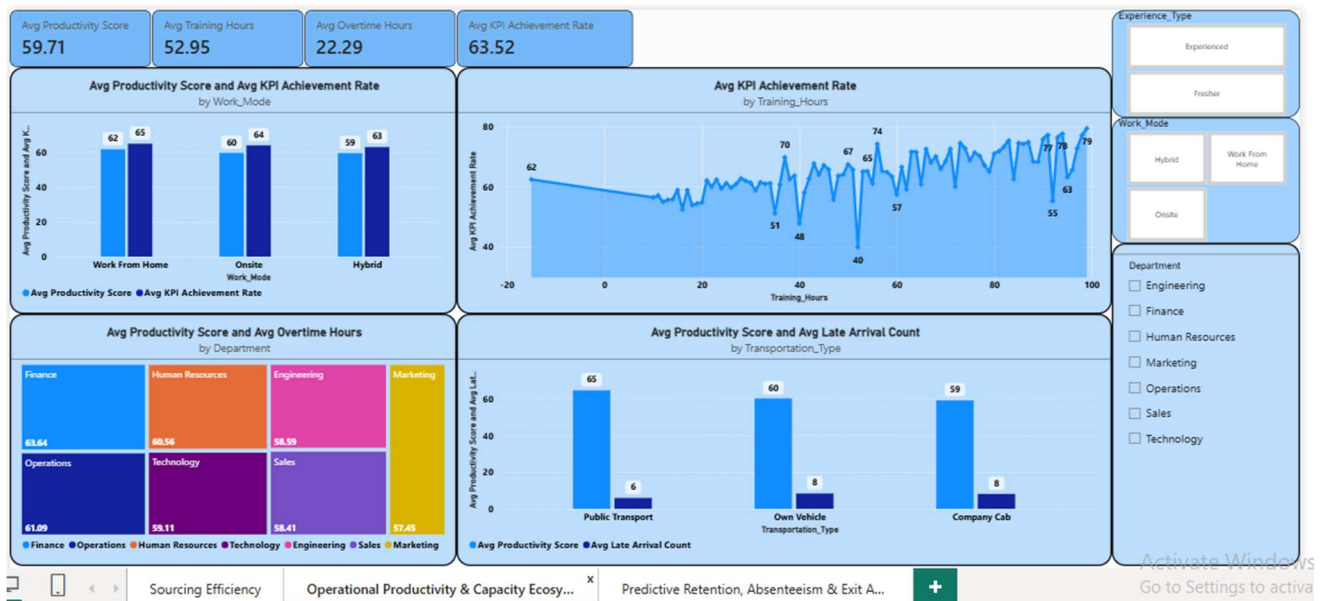
- 1.Total Headcount
 - 2.Baseline Attrition Rate (%)
 - 3.Average Cost Per Hire
 - 4.Average Productivity Score
 - 5.High Flight Risk Count and etc..
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8. Analysis and Visualizations (Power BI)



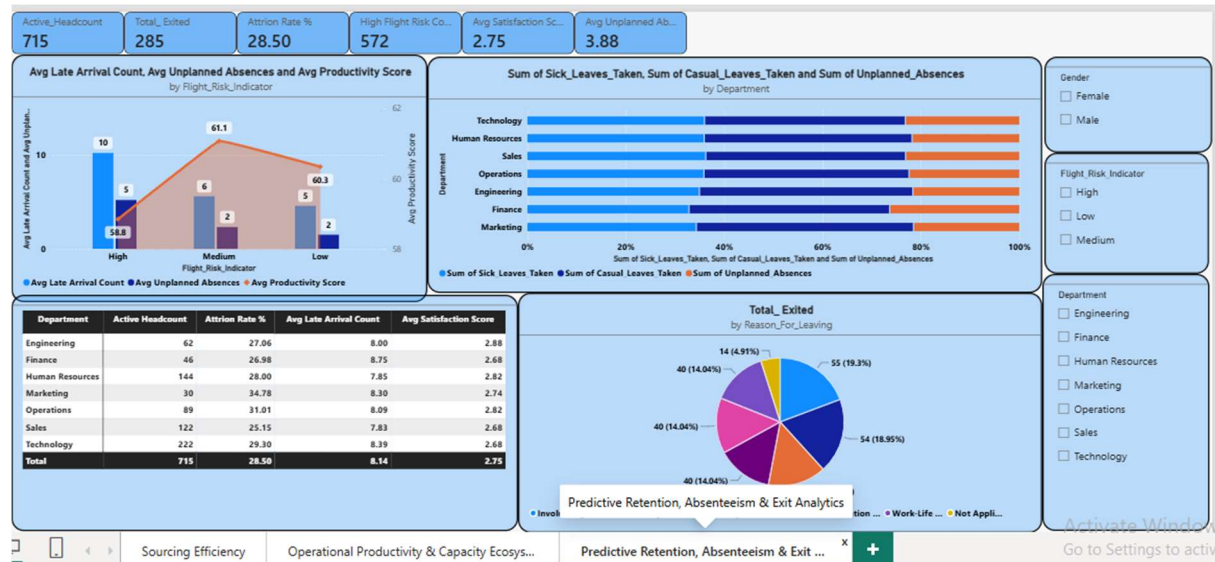
Dashboard 1: Sourcing Efficiency

- KPI Cards: Total Hires (1,000) Total Hiring Cost (₹14,45,000), Avg Cost Per Hire (₹1,445), Avg Satisfaction Score (2.75), High Risk Rate % (57.20%), Avg Productivity Score (59.7), Channel Attrition Rate % (28.50%).
- Bar Visual: Total Hires by Source_Of_Hiring & Experience_Type (Naukri, LinkedIn, Company Career Site, Walk-In, Referral, Campus).
- Donut Chart: Gender Distribution (63.8% Male, 36.2% Female).
- Combo Chart: Avg Productivity Score, KPI Achievement Rate, and High Risk Rate % across sourcing channels.
- Scatter Plot: Cost Per Hire vs. Attrition Rate % across channels.
- Interactive Slicers: Region, Department, Experience Type.



Dashboard 2: Operational Productivity & Capacity Ecosystems

- KPI Cards: Avg Productivity (59.71), Avg Training Hours (52.95), Avg Overtime Hours (22.29), Avg KPI Achievement Rate (63.52%).
- Clustered Column Chart: Productivity vs. KPI Achievement by Work_Mode (WFH, Onsite, Hybrid).
- Line Chart: Avg KPI Achievement Rate vs. Training_Hours.
- Treemap: Productivity Score and Overtime Hours by Department.
- Bar Chart: Productivity vs. Late Arrival Count by Transportation_Type.



Dashboard 3: Predictive Retention, Absenteeism & Exit Analytics

- KPI Cards: Active Headcount (\$715\$), Exited Count (\$285\$), Attrition Rate (\$28.50\%\$), High Flight Risk Count (\$572\$), Avg Unplanned Absences (\$3.88\$).
- Combo Visual: Late Arrivals, Unplanned Absences, and Productivity Score by Flight_Risk_Indicator (High, Medium, Low).
- 100% Stacked Bar Chart: Sick Leaves, Casual Leaves, and Unplanned Absences distribution across Departments.
- Summary Table: Departmental Breakdown (Active Headcount, Attrition Rate %, Late Arrival Count, Satisfaction Score).
- Pie Chart: Total Exited distribution by Reason_For_Leaving

9. Insights & Conclusions

Key Findings: Summarize trends, patterns, or anomalies identified in the data

Descriptive Analysis (What happened?)

- The organization hired 1,000 total employees across major channels including Naukri (217 hires), LinkedIn (183 hires), and Company Career Site (165 hires) at an average cost per hire of ₹1,445.
- The current active workforce stands at 715 employees, with 285 total exits, resulting in an overall baseline attrition rate of 28.50%.
- Average productivity score across departments stays steady at 59.71, while training hours average 52.95 hours per employee

2. Diagnostic Analysis (Why did it happen?)

- High flight risk employees exhibit distinct behavioral anomalies: they log higher average late arrivals (10 count) and unplanned absences (5 count) compared to low flight risk peers (5 late arrivals, 2 unplanned absences).
- Departmental workload and culture variations contribute to turnover disparities—Marketing faces the highest attrition at 34.78%, whereas Sales retains workforce stability with a lower attrition rate of 25.15%.
- Lower KPI achievement rates (40–50%) correlate directly with low training exposure (≤ 30 hours), proving that skill gaps directly suppress performance output.

3. Predictive Analysis (What is likely to happen?)

- Out of 715 active employees, 572 (80%) show indicators placing them in the High Flight Risk category, signaling potential future turnover if unaddressed.
- Employees accumulating more than 8 late arrivals and 4 unplanned absences per quarter are at the highest probability of initiating an exit within 3–6 months.
- Sustained training programs exceeding 70+ hours predictably boost KPI achievement rates to the 70–79% range

4 Prescriptive Analysis (What action should we take?)

- Reallocate talent acquisition budget toward cost-effective channels (Naukri, LinkedIn) and restructure the Employee Referral program to improve post-hire retention.
- Implement an automated early-warning tracking system for attendance anomalies (≥ 8 late arrivals, ≥ 4 unplanned absences) to trigger HR stay-interviews before resignation.

- **Mandate a minimum threshold of 50+ training hours annually per employee to uplift baseline KPI achievement and mitigate flight risk in high-turnover units like Marketing and Operations.**
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10. Conclusions

The findings show that regular training directly improves performance, while attendance anomalies like late arrivals serve as early warning signs for employee turnover. Using these interactive dashboards enables management to shift from reactive exit tracking to proactive retention, optimizing recruitment spend and improving overall workforce efficiency..